

From Fiori Save to My Inbox

Building an End-to-End Travel Approval App with SAP CAP, HANA Cloud, and SAP Build Process Automation

SAP Community article draft + LinkedIn launch copy

Publishing package

Recommended SAP Community title

From Fiori Save to My Inbox: An End-to-End Travel Approval App with SAP CAP, HANA Cloud, and SAP Build Process Automation

Short description / teaser

This hands-on guide builds a Travel Approval application from the ground up on SAP BTP. A Fiori Elements app saves travel requests in SAP HANA Cloud, CAP automatically starts an SAP Build Process Automation workflow, an approver acts from SAP Build Lobby->My Inbox, and the decision is written back to the same Travel record. The guide includes the CAP files, Cloud Foundry commands, HDI binding, EDMX-based Actions, BPA configuration, API-key setup, testing, monitoring, and the troubleshooting lessons that made the final flow work.

Suggested SAP Community tags

SAP Cloud Application Programming Model | SAP Business Technology Platform | SAP HANA Cloud | SAP Business Application Studio | SAP Build Process Automation | Fiori Elements | OData V4

LinkedIn headline

Built on SAP BTP: Fiori + CAP + HANA Cloud + SAP Build Process Automation, working end-to-end

LinkedIn post - recommended

I just completed an end-to-end Travel Approval application on SAP BTP. A travel request is created in a Fiori Elements app, persisted in SAP HANA Cloud through CAP, and automatically sent to SAP Build Process Automation. The approver receives the task in My Inbox, approves or rejects it, and the final status is written back to the Travel record.

The most valuable part was not only getting the happy path to work. It was understanding what broke along the way: mismatched HDI containers, stopped Cloud Foundry instances, URL-prefix issues in Actions, draft activation behavior, technical-user task assignment, and the difference between CAP XSUAA credentials and BPA service-key credentials.

I documented the complete build, deployment, testing, monitoring, and troubleshooting sequence so others can reproduce the flow with less guesswork.

#SAPCommunity #SAPBTP #SAPCAP #SAPHANACloud #SAPBuild #ProcessAutomation #FioriElements #OData

LinkedIn post - concise alternative

New hands-on SAP BTP build: a Travel Approval app using Fiori Elements, SAP CAP, HANA Cloud, and SAP Build Process Automation.

Create a Travel request in Fiori -> CAP saves it in HANA -> BPA starts automatically -> the task appears in My Inbox -> Approve or Reject -> the status returns to the Travel record.

The write-up includes the files, terminal commands, Actions, API trigger, API key, monitoring, and the fixes for the real issues encountered along the way.

#SAPBTP #SAPCAP #SAPBuild #SAPHANACloud #FioriElements

SAP Community blog draft

Goal

The goal is to build a working Travel Approval application, not just a collection of disconnected demos. A user creates a travel request in a Fiori Elements application. The CAP service stores the request in SAP HANA Cloud and automatically starts an SAP Build Process Automation workflow.

An approver opens the task in My Inbox, chooses Approve or Reject, and the final decision updates the same Travel record in the HANA Cloud.

```

Create Travel in Fiori
↓
CAP saves the active record in HANA Cloud
↓
CAP starts the BPA workflow automatically
↓
Approval task appears in My Inbox
↓
Approve → overall_status = A
Reject → overall_status = R
↓
Validate in Fiori, Monitoring, and HANA Database Explorer
    
```

In plain English: CAP is the application and API layer, HANA Cloud stores the data, Fiori is the user interface, and SAP Build Process Automation manages the human approval step.

Prerequisites

BTP services and instances used

Service / instance	Plan or type	Why it is needed
SAP Business Application Studio	Subscription / development environment	Create and run the CAP and Fiori projects.
SAP HANA Cloud	HANA database	Persistent database used by the CAP application.
CAPWF-db	HDI container, hdi-shared	Application-owned schema for Travel data.
CAPWF-auth	XSUAA, application	Protect the deployed CAP service.
CAPWF-destination-service	Destination, lite	Destination content and runtime connectivity.
CAPWF-html5-service	HTML5 Application Repository,	Host the deployed UI content

	app-host	when required.
BPA	process-automation-service	Create, deploy, trigger, and monitor the approval process.
CAPWF-srv	Cloud Foundry application	Runs the deployed CAP service.

Role collections used for this scenario

Area	Role collection	Use in this guide
SAP Business Application Studio	Business_Application_Studio_Developer	Open a dev space and develop the CAP/Fiori application.
SAP Build Process Automation	ProcessAutomationDeveloper	Create, edit, release, and publish actions and process artifacts.
SAP Build Process Automation	ProcessAutomationParticipant	Open and complete assigned tasks in My Inbox.
SAP Build Process Automation	ProcessAutomationAdmin	Administer process configuration and authorizations where required.
BTP subaccount setup	Subaccount Administrator or delegated service/destination admin roles	Create service instances, destinations, role assignments, and service keys.
SAP HANA Cloud	Tenant-specific HANA Cloud administration permissions	Create/start the HANA database and administer database access.

Role note: Exact role assignments can vary by tenant setup and organizational policy. Use the least privilege and confirm the current SAP Help documentation for your account.

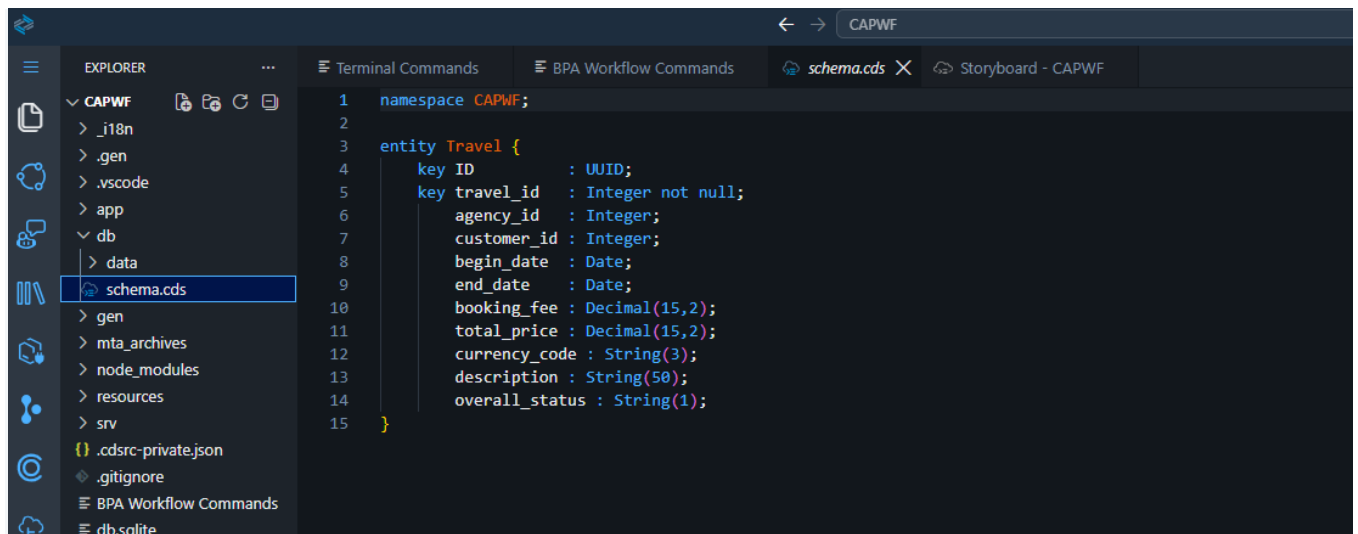
The End-to-End Flow

1. Model the Travel entity in CDS.
2. Expose the entity as a draft-enabled OData V4 service.
3. Generate the Fiori Elements application.
4. Bind the CAP project to the HANA HDI container.
5. Build the MTA archive and deploy it to Cloud Foundry.
6. Download the CAP metadata as an EDMX specification.

7. Create SAP Build Actions from the EDMX and test the OData operations.
8. Create the Travel Context, API trigger, Approval Form, and approval of branches.
9. Create a BPA service key and an API key with trigger execution permission.
10. Add the CAP SAVE handler that starts the workflow automatically.
11. Test from Fiori, process the task in My Inbox, monitor the instance, and validate the final status in HANA.

1. Model the CDS Entity for the CAP OData V4 Service

The CDS entity is the source of truth for the Travel data. The technical UUID and the business Travel ID form the key used by the draft-enabled service and by the generated SAP Build Actions.



You can use Using BAS Productivity Tools

Status values: W means waiting for approval, A means approved, and R means rejected.

1.1 Expose the Travel entity as a service

Create file `srv/service.cds`.

```

1 using { CAPWF as my } from '../db/schema.cds';
2
3 @path:'/service/CAPWFSERVICE'
4 service CAPWFSERVICE {
5
6   @odata.draft.enabled
7   entity Travel as projection on my.Travel actions {
8
9     action submitForApproval();
10
11   };
12
13 }
14
15 //annotate CAPWFSERVICE with @requires: ['authenticated-user'];
16

```

Why draft support matters: Fiori Elements creates and edits a draft first. The active Travel record is produced when the user activates or saves the draft. The CAP SAVE event is therefore the reliable point for starting the workflow.

2. Create the Fiori Elements App

Generate a List Report/Object Page application for CAPWFSERVICE.Travel. The list lets users create requests and review the current status. The Object Page captures the full Travel details.

Default sort by Travel ID

```

annotate CAPWFSERVICE.Travel with @(
  UI.PresentationVariant : {
    Visualizations : ['@UI.LineItem'],
    SortOrder : [
      { Property : travel_id, Descending : false }
    ]
  }
);

```

Lesson learned: SortOrder did not apply consistently until the Presentation Variant explicitly referenced @UI.LineItem through Visualizations.

<

SAP

Preview – List of CAPWFSERVICE.Travel

DU

Standard

No filters active

(15)

CreateDelete

Trav...	Agency ID	Customer ID	Overall Sta...	Booking Fee	Total Price	Currency ...	Begin Date	End Date	Travel UUID	Description	
☐	1	70,001	10,001	O	50.00	1,200.00 USD	Oct 1, 2026	Oct 5, 2026	11111111-1111-1111-1111-111111111111	London Customer Visit	>
☐	2	70,002	10,002	O	75.00	2,500.00 USD	Nov 10, 2026	Nov 15, 2026	22222222-2222-2222-2222-222222222222	Instances up	>
☐	3	70,003	10,003	A	25.00	500.00 EUR	Oct 20, 2026	Oct 22, 2026	33333333-3333-3333-3333-333333333333	SAP CAP Training	>
☐	4	70,004	10,004	X	30.00	950.00 USD	Dec 1, 2026	Dec 3, 2026	44444444-4444-4444-4444-444444444444	Workshop in New York	>
☐	5	70,005	10,005	O	40.00	1,500.00 GBP	Oct 15, 2026	Oct 18, 2026	55555555-5555-5555-5555-555555555555	Project Kickoff Meeting	>

Figure 1. Fiori List Report with Travel IDs in ascending order.

3. Connect to Cloud Foundry

Cloud Foundry provides the runtime for the deployed CAP service. Log in from the BAS terminal and select the correct organization and dev space.

```
cf login
cf target
```

Before testing SAP Build Actions, always confirm that the deployed CAP application is running:

```
cf start CAPWF-srv
cf app CAPWF-srv
```

Expected status:

```
requested state: started
instances: 1/1
state: running
```

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  SQL CONSOLE

user: CAPWF $ cf start CAPWF-srv
last uploaded:   Wed 30 Sep 22:33:32 UTC 2026
stack:           cflinuxfs4
buildpacks:
isolation segment: trial
                  name      version  detect output  buildpack name
                  nodejs_buildpack  1.9.4   nodejs         nodejs

type:            web
sidecars:
instances:       1/1
memory usage:    1024M
state            since              cpu    memory    disk    logging    cpu entitlement  details  ready
#0  running      2026-10-05T22:03:59Z  0.0%   0B of 0B   0B of 0B  0B/s of 0B/s  0.0%           details  true
user: CAPWF $
connect

```

Important: When CAPWF-srv was stopped, every SAP Build Action returned 404 Requested route does not exist. Starting the application restored the Actions immediately.

4. Bind to a HANA Cloud HDI Container

An HDI container is an application-scoped database container. It gives the CAP application its own isolated schema inside the HANA Cloud database.

```
cds bind --to CAPWF-db
cds env get requires.db --profile hybrid
```

The hybrid profile must show the CAPWF-db instance. Start the local service against HANA:

```
pkill -f cds
cds watch --profile hybrid
```

Look for connect to db > hana and the expected HANA schema in the terminal output.

Lesson learned: The lowercase capwf-db and uppercase CAPWF-db were two separate service instances. Local Fiori and deployed BPA initially showed different record counts because they were connected to different HDI containers.

5. Generate the MTAR File

The Multi-Target Application archive packages the service, database deployer, security resources, destinations, and HTML5 content into one deployable unit.

```
mbt build
```

The archive is generated under:

```
mta_archives/CAPWF_1.0.0.mtar
```

6. Deploy to HANA Cloud and Cloud Foundry

```
cf deploy mta_archives/CAPWF_1.0.0.mtar
```

The deployment runs the database deployer, binds CAPWF-srv to CAPWF-db and CAPWF-auth, and starts the application route.

```
cf app CAPWF-srv
cf routes
cf logs CAPWF-srv --recent
```

Free-tier note: Stopping applications frees runtime memory, but it also makes the CAP route unavailable. Start CAPWF-srv again before testing browser URLs or SAP Build Actions.

7. Connect HANA Database Explorer and Add the HDI Container

1. Open SAP HANA Database Explorer.
2. Choose Add Database or Add HDI Container.
3. Select the CAPWF-db HDI container.
4. Expand Catalog and Tables.
5. Open the Travel table data and compare the record count with the Fiori app.
6. After approval or rejection, refresh the table and verify overall_status.

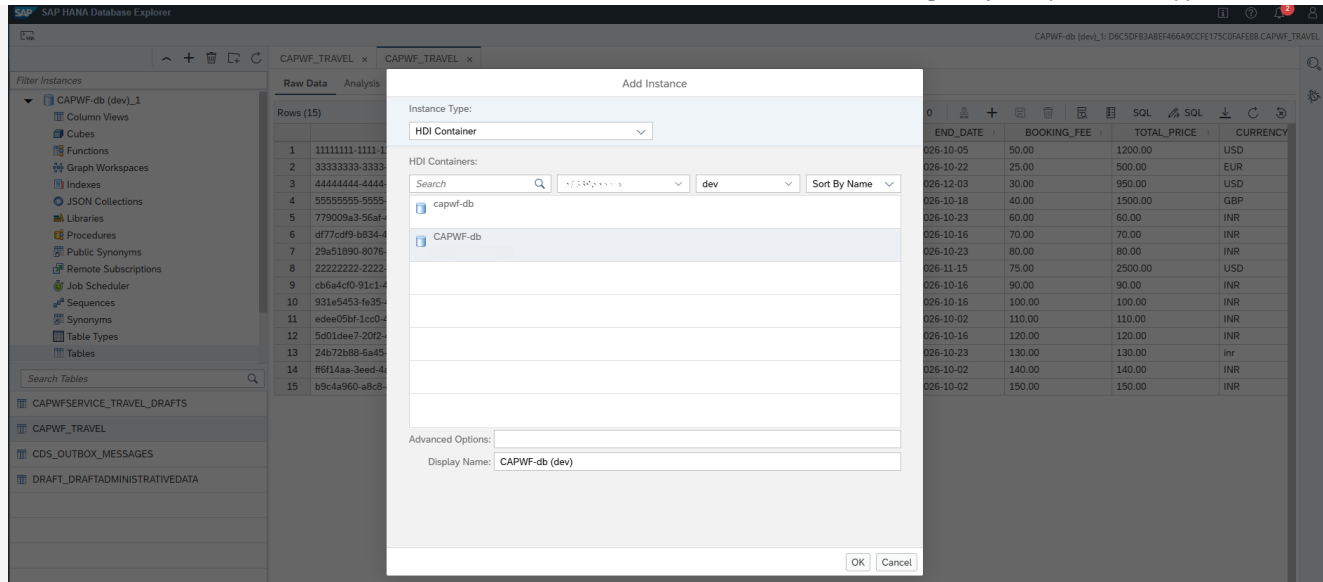


Figure 2. CAPWF-db added to SAP HANA Database Explorer.

8. Download the Metadata File as EDMX

The metadata document describes the OData V4 service, entity fields, keys, draft properties, and custom actions. SAP Build uses this specification to generate reusable Actions.

```
/service/CAPWFService/$metadata
```

Open the metadata->right-click->Save as .edmx file

Open the local BAS port 4004 metadata URL, save the XML as CAPWFService.edmx, and verify that it contains:

- EntitySet Name="Travel"
- The Travel entity fields and keys
- Draft operations
- Action Name="submitForApproval" IsBound="true"

9. Create SAP Build Actions

1. Open SAP Build Lobby -> Actions -> Create.
2. Choose Upload API Specification.
3. Upload CAPWFService.edmx.
4. Create the Action project CAPWFService3.
5. Select the required Travel operations. For POC, I have selected all 13 generated operations.
6. Open Settings and configure the destination and URL prefix.

Setting	Value
---------	-------

Destination	CAPWF-srv
URL Prefix	/service/CAPWFService
CSRF	Disabled for this POC
Destination authentication	OAuth2ClientCredentials

Test the Actions before publishing

- Get entities from Travel should return 200 OK and all records. - see below image
- Get entity from Travel by key should return the selected record using ID, travel_id, and IsActiveEntity=true.
- Update entity in Travel should be validated independently before adding it to the process.
- Release and Publish only after direct tests succeed.

Troubleshooting shortcut: If all Actions suddenly return 404 and no request reaches, run the below commands in new Terminal.

1. cf logs CAPWF-srv
2. cf app CAPWF-srv
3. cf start CAPWF-srv -->You can see the Instances are up and Running 1/1

The root cause may simply be a stopped application.

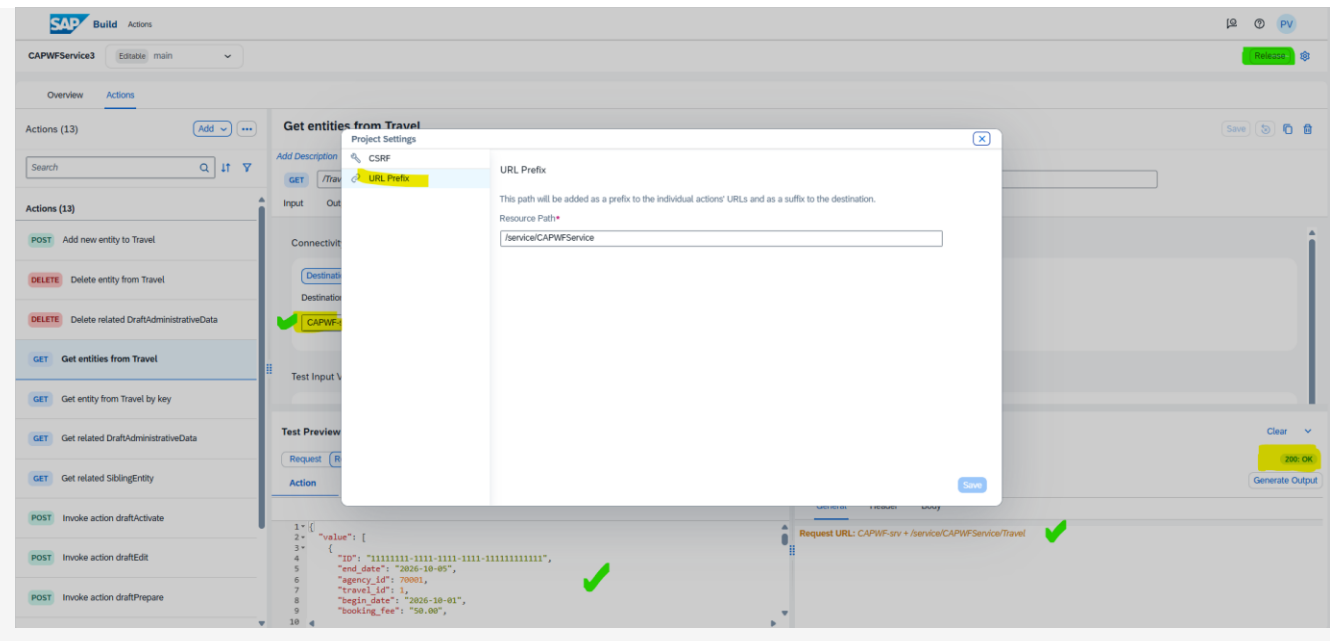


Figure 3. CAPWFService3 Action test returning 200 OK.

10. Steps in SAP Build Process Automation

Create the project and artifacts

Secrets removed | Replace screenshot placeholders before publishing

1. Create the project RAP110_Travel_Workflow.
2. Create the process of Travel Approval Processing.
3. Create the Travel Context datatype.
4. Create a Travel Approval Trigger as an API trigger.
5. Create the Approval Form and map the Travel fields.
6. Add Approve and Reject outcomes.
7. Add Update entity in Travel from the published CAPWFSERVICE3 dependency.
8. Map the keys and set overall_status to A in Approve and R in Reject.

The screenshot displays the SAP Build Process Automation interface. The top section shows the project overview for 'RAP110_Travel_Workflow' (version 8.0.0, Deployed, Active). It includes creation details (Created On: October 2, 2026, By: You) and a list of artifacts: Approval Form, Travel Approval Processing, and Travel Context. Below this, a table lists the artifacts with their descriptions and types.

Name	Description	Type
Approval Form	Form to approve or reject Travel	Form
Travel Approval Processing	Process to handle Travel Workflow	Process
Travel Context	Travel Context	Data Type

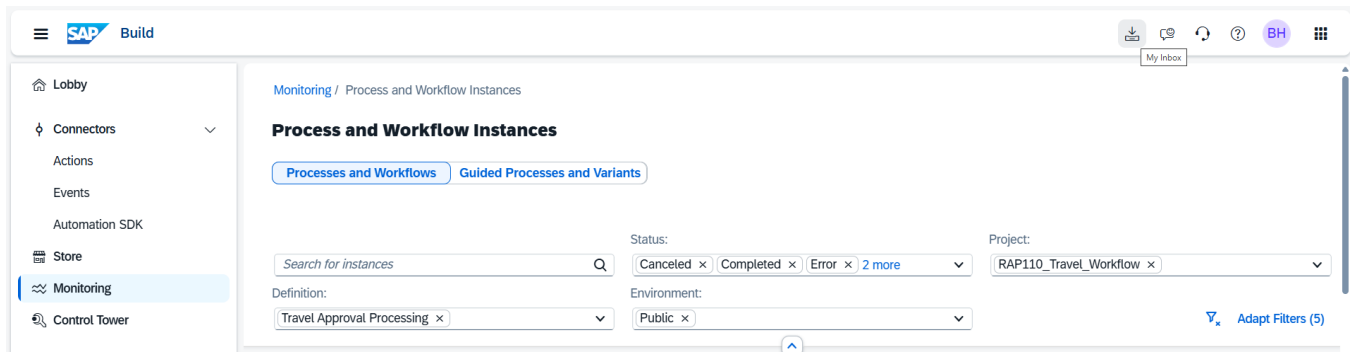
The bottom section shows the 'Approval Form' design. It includes a header 'Travel Approval' and a message: 'A new travel request has been received. Please review and confirm whether it can be fulfilled or not.' Below this, the 'Travel Details' section lists various fields for input, including Travel ID, Agency ID, Customer ID, Overall Status, Currency Code, Booking Fee, Total Price, End Date, Begin Date, and Description.

Travel Context fields

RAP110_Travel_Workflow				
Name	Type	Sample	List	Required
TravelUUID	String		No	No
TravelID	Number		No	No
IsActiveEntity	Boolean		No	No
agency_id	Number		No	No
customer_id	Number		No	No
begin_date	Date		No	No
end_date	Date		No	No
booking_fee	Number		No	No
total_price	Number		No	No
currency_code	String		No	No
description	String		No	No
overall_status	String		No	No

Approval Form recipient

When a human starts the workflow manually, Started By is the signed-in user. When CAP starts the process using client credentials, Started By is a technical client. For the POC, the Approval Form recipient was set to the intended email address, so the work item appeared in My Inbox. A production design should pass an Approver Email value or determine the approver through business rules.



11. Add and Delete Dependencies

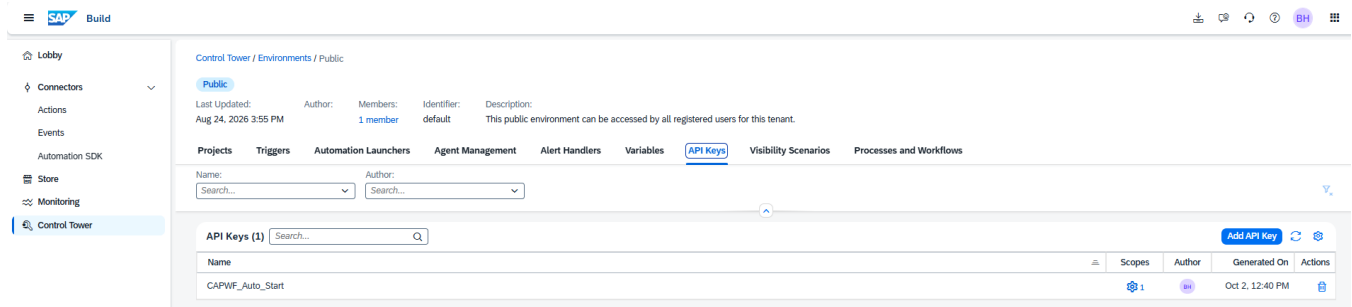
A process can be used only when released and published Action projects through Dependencies.

1. Open the editable BPA project and switch to Dependencies.
2. Add the published CAPWFSservice3 Action project.
3. If an old Action dependency is removed, delete any orphaned process nodes that show blank Inputs.
4. Re-add the Actions from CAPWFSservice3 and remap their inputs.
5. To remove a dependency, select it, choose Settings, and select Delete.
6. If an Action project cannot be deleted, remove consuming dependencies and delete its released versions first.

12. Create an API Key in SAP Build

The CAP application starts the process as a technical client. It therefore needs both OAuth client credentials and an SAP Build API key that is authorized to execute the trigger.

1. Open SAP Build -> Control Tower -> API Keys, or Public Environment -> API Keys.
2. Choose Add API Key.
3. Enter a meaningful name, such as CAPWF_Auto_Start.
4. Select the Public environment.
5. Enable trigger_execute. This is the minimum scope needed to execute the API trigger.
6. Create the key and copy it immediately. The value is displayed only once.



13. Manual Workflow Test Using the Payload

Before automating the trigger from CAP, manually prove the deployed process and payload schema. See image

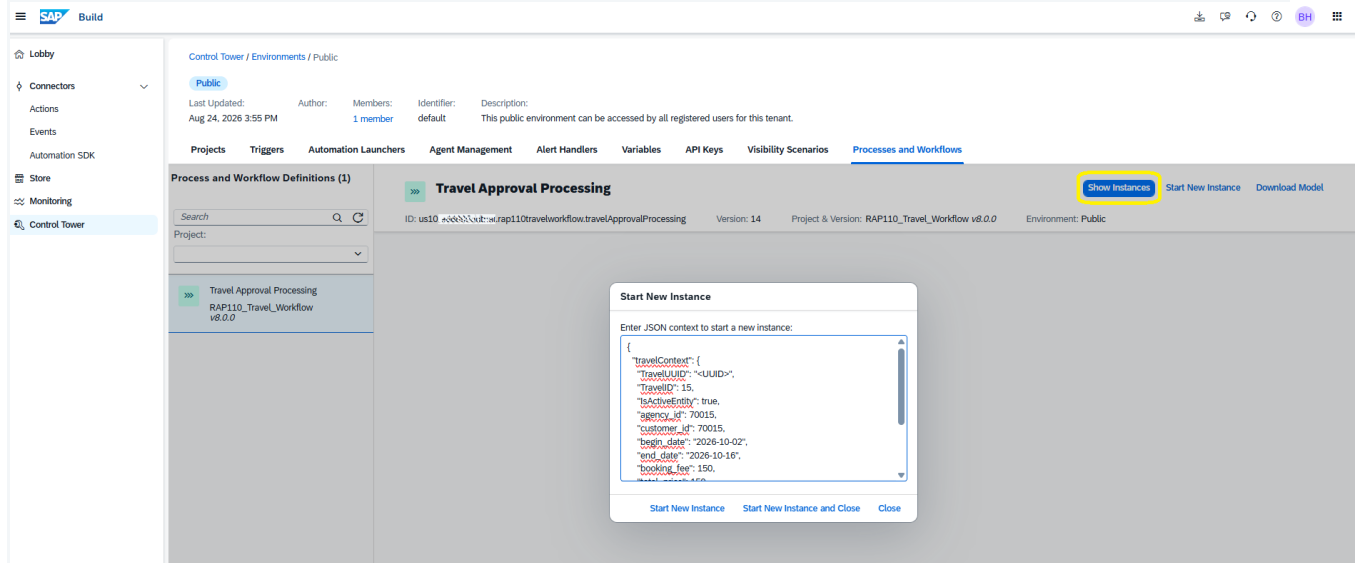
1. Open Control Tower -> Environments -> Public -> Triggers.
2. Open Travel Approval Trigger.
3. Choose Start New Instance.
4. Paste only the process context expected by the dialog, with travelContext as the root object.
5. Start the instance and confirm it appears in Monitoring.
6. Open My Inbox and process the task.
7. Confirm the instance completes, and the Travel status is updated.

```
{
  "travelContext": {
    "TravelUUID": "<UUID>",
    "TravelID": 15,
    "IsActiveEntity": true,
    "agency_id": 70015,
    "customer_id": 70015,
    "begin_date": "2026-10-02",
    "end_date": "2026-10-16",
```

```

"booking_fee": 150,
"total_price": 150,
"currency_code": "INR",
"description": "Travel request",
"overall_status": "W"
}
}

```



14. List of Terminal Commands to Get BPA Keys

```

cf create-service-key BPA cap-bpa-key
cf service-key BPA cap-bpa-key

```

From the BPA service key, record these values securely: clientid, clientsecret, and url. The token endpoint is the url value plus /oauth/token.

Do not mix credentials: CAPWF-auth credentials protect the CAP application. BPA service-key credentials authenticate the call to SAP Build Process Automation. Using CAPWF-auth credentials caused 401 invalid_client / Bad credentials.

15. Start the Workflow Automatically from CAP

The Fiori application activates a draft when the user saves a new Travel. CAP handles that lifecycle through the SAVE event. The handler builds the exact trigger payload, obtains an OAuth token, sends the BPA API key, and creates the workflow instance.

```

// srv/service.js - relevant pattern
this.after('SAVE', 'Travel', async (data, req) => {
  const travel = req.data;

  const workflowPayload = {

```

```

definitionId:
  'us10.e66895ddtrial.rap110travelworkflow.travelApprovalProcessing',
context: {
  travelContext: {
    TravelUUID: travel.ID,
    TravelID: travel.travel_id,
    IsActiveEntity: true,
    agency_id: travel.agency_id,
    customer_id: travel.customer_id,
    begin_date: travel.begin_date,
    end_date: travel.end_date,
    booking_fee: Number(travel.booking_fee),
    total_price: Number(travel.total_price),
    currency_code: travel.currency_code,
    description: travel.description,
    overall_status: travel.overall_status
  }
}
};

// 1. Get an OAuth token with BPA client credentials.
// 2. POST workflowPayload to BPA_WORKFLOW_URL.
// 3. Send Authorization: Bearer <token> and api-key headers.
});

```

Production recommendation: Add idempotency. Store the workflow instance ID or a submitted flag so editing the same Travel later does not start a duplicate approval process.

16. Verify in BAS with Three Terminals

Terminal 1 - Run local CAP with BPA credentials

```

export BPA_CLIENT_ID='<BPA service-key clientid>'
export BPA_CLIENT_SECRET='<BPA service-key clientsecret>'
export BPA_TOKEN_URL='<BPA service-key url>/oauth/token'
export BPA_API_KEY='<SAP Build Public environment API key>'
export BPA_WORKFLOW_URL='https://spa-api-gateway-bpi-us-
prod.cfapps.us10.hana.ondemand.com/workflow/rest/v1/workflow-instances'

pkill -f cds
cds watch --profile hybrid

```

Terminal 2 - Confirm the deployed CAP app is running

```
cf start CAPWF-srv  
cf app CAPWF-srv
```

Terminal 3 - Watch deployed CAP requests

```
cf logs CAPWF-srv
```

A successful automatic start prints:

```
BPA workflow started for Travel 15  
Workflow instance ID: <generated instance ID>
```

Session behavior: The export commands apply only to the current terminal session. After BAS signs out or a new terminal is opened, export the variables again before `cds watch`.

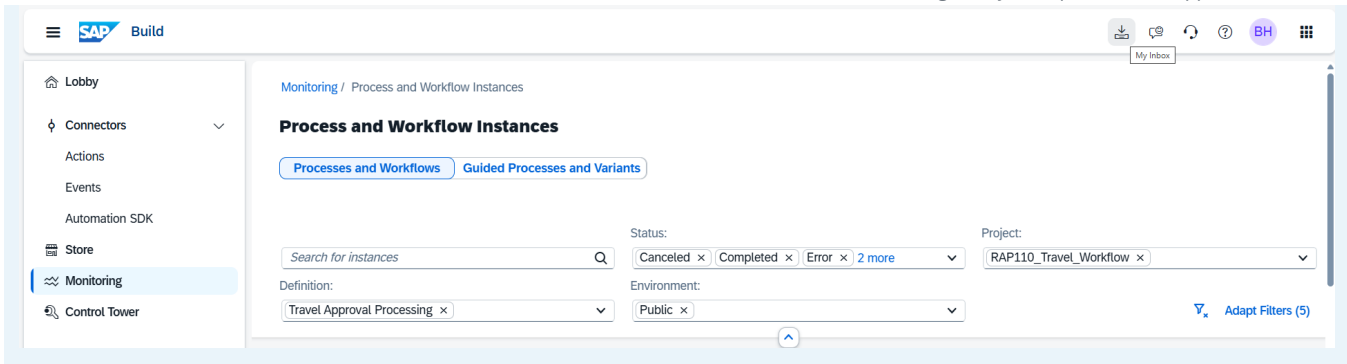
17. Test Directly from the Fiori Application

1. Confirm Terminal 1 has the BPA environment variables, and `cds watch` is running.
2. Confirm that CAPWF-srv is 1/1 running if the workflow calls CAP through SAP Build Actions.
3. Open the Fiori Preview.
4. Create a new Travel record with `overall_status W`.
5. Save the record.
6. Watch Terminal 1 for the workflow instance ID.
7. Open SAP Build My Inbox and process the Travel task.
8. Refresh the Fiori List Report and confirm A or R.

18. My Inbox - Approve or Reject

The approver opens the Travel task, reviews the Travel fields, and chooses Approve or Reject. The process follows the corresponding branch and calls the CAP update Action. The completed workflow should disappear from My Inbox and appear as Completed in Monitoring.

If My Inbox is empty: Open the instance Logs. If the recipient is a technical client instead of the intended user, revise the Approval Form recipient for mapping and deploying a new version.



19. Monitor Workflow Instances

1. Open SAP Build -> Monitoring -> Processes and Workflows.
2. Open Travel Approval Processing.
3. Filter by Running, Error, or Completed.
4. Open an instance and inspect Logs to see each executed step.
5. Open Context to confirm startEvent.travelContext values.
6. Use the instance ID printed by CAP to correlate the application request with the workflow instance.

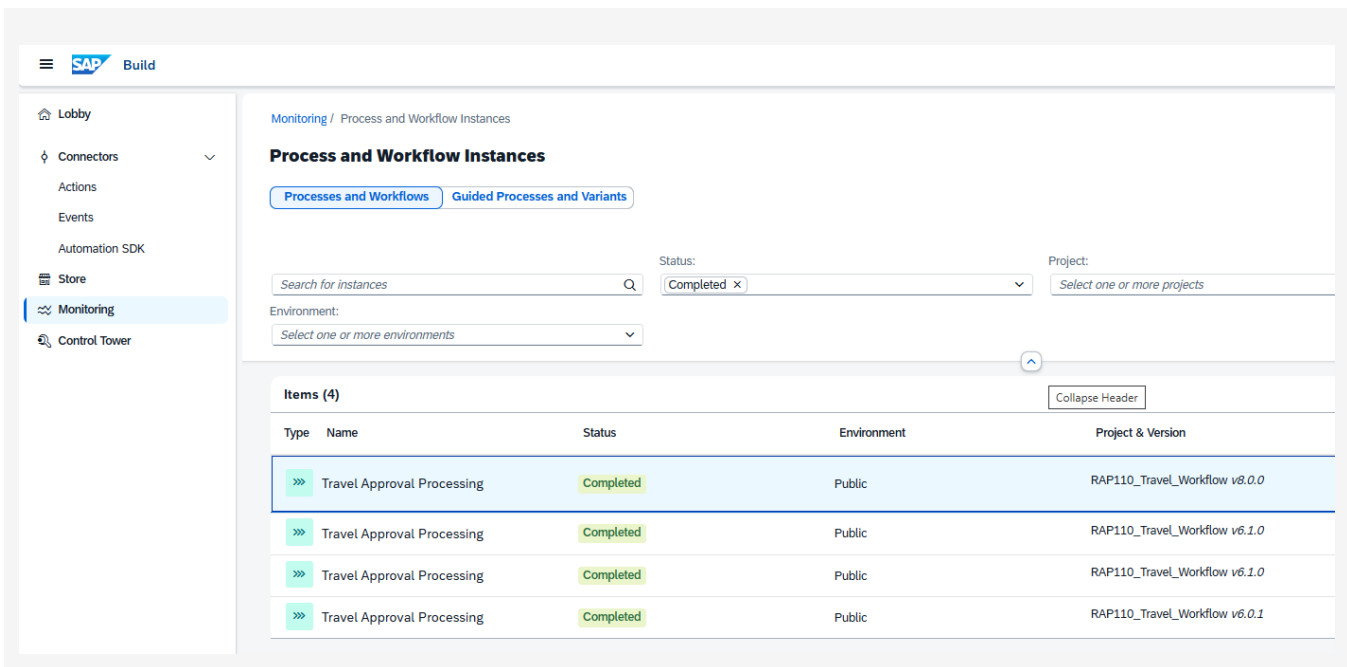


Figure 4. Completed Travel Approval Processing instance.

20. Validate in the HDI Container

1. Open SAP HANA Database Explorer and refresh CAPWF-db.
2. Open the Travel table for data.
3. Locate the Travel ID used in the test.

4. Confirm W before approval.
5. Confirm 'A' after approval or 'R' after rejection.
6. Compare the results with the Fiori List Report.

What Broke, Why It Broke, and the Fix

Symptom	Root cause	Fix
Fiori showed more records than BPA/HANA	Different HDI containers were used	Bind local hybrid runtime to CAPWF-db.
Port 4004 already in use	An old cds process was still running	pkill -f cds, then start one runtime.
Actions returned 404 route does not exist	CAPWF-srv was stopped	cf start CAPWF-srv and confirm 1/1 running.
Metadata returned Unauthorized	XSUAA was protecting the deployed service	Use OAuth2ClientCredentials in CAPWF-srv destination.
BPA trigger returned invalid_client	CAPWF-auth credentials were used instead of BPA service-key credentials	Use cf service-key BPA cap-bpa-key values.
Workflow URL was Invalid URL	Environment variables were missing after BAS re-login	Export variables in the same terminal before cds watch.
Workflow instance existed but My Inbox was empty	Started By resolved to the technical OAuth client	Use an explicit recipient or dynamic ApproverEmail.
Action Inputs were blank	Process node referenced a removed Action dependency	Delete and re-add the node from the active dependency.
Travel ID sorting was inconsistent	Presentation Variant did not reference the LineItem visualization	Add Visualizations: ['@UI.LineItem'].
SAVE handler received data differently than CREATE	Fiori used draft activation	Use after SAVE and req.data.

Conclusion

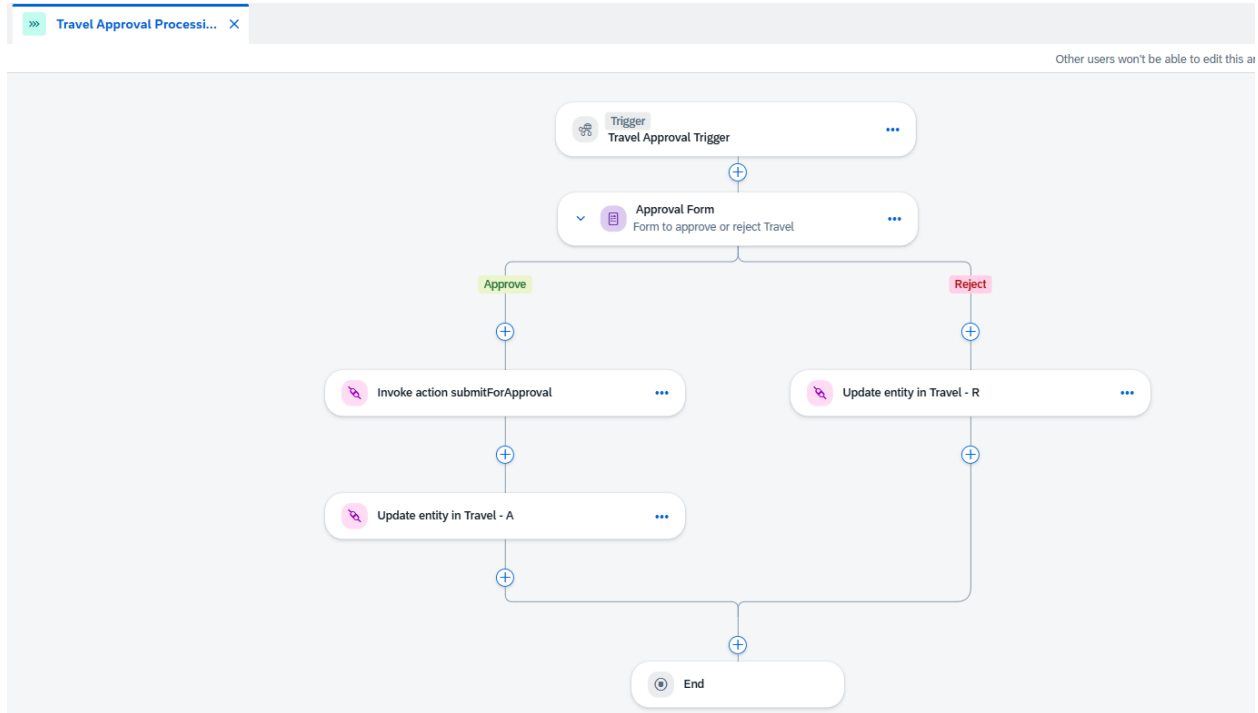
The final solution connects four SAP BTP capabilities into one coherent business flow: Fiori Elements for data entry, CAP for service logic, HANA Cloud for persistence, and SAP Build Process Automation for human approval. The biggest lesson was that end-to-end success depends on more than code. Runtime state, destinations, service keys, API keys, task recipients, action dependencies, and the exact HDI binding all matter. Once those pieces were aligned, the flow became predictable: create the Travel record, receive the task, approve or reject, and see the result in the same application.

If you are building a similar integration, validate each boundary independently: Fiori to CAP, CAP to HANA, CAP to BPA, BPA to CAP Actions, and BPA task assignment. That approach turns a complex landscape into a series of small, testable checkpoints.

Screenshot plan for the published blog

Use screenshots at the points below. Crop out email addresses, client IDs, secrets, API keys, database credentials, and tenant-specific identifiers before publishing.

Placement	Screenshot
Prerequisites	BTP Instances and Subscriptions page
CAP model	BAS project tree and schema.cds
HANA binding	Terminal showing connect to db > hana
Database Explorer	CAPWF-db and Travel data
Fiori	Sorted Travel List Report
Metadata	Browser metadata with Travel EntitySet
Actions	CAPWFService3 Settings and 200 OK response
Process	Travel Approval Processing canvas
Context	Travel Context datatype and mappings
API Key	API Key setup screen with key value hidden
Auto trigger	Terminal showing workflow instance ID
Inbox	Travel approval task
Monitoring	Completed instance Logs and Context
Validation	HANA/Fiori status A or R



Available screenshot from the implementation session. Replace or recaption as appropriate before publishing.

References

- **SAP Help: SAP Build Process Automation authorizations:** https://help.sap.com/docs/PROCESS_AUTOMATION/a331c4ef0a9d48a89c779fd449c022e7/d65afd67cd15407da0f2cf4a3b1b3f6f.html?locale=en-US
- **SAP Help: SAP Business Application Studio roles:** <https://help.sap.com/docs/bas/sap-business-application-studio/manage-authorizations-and-roles>
- **SAP Help: API Keys:** <https://help.sap.com/docs/build-process-automation/sap-build-process-automation/api-keys>
- **SAP Help: Execute an API Trigger:** <https://help.sap.com/docs/build-process-automation/sap-build-process-automation/execute-api-trigger-c6dd6c22a0f84ed5aa33fa24284acbc1?locale=en-US&mt=ja-JP>
- **SAP Help: Use API Keys:** <https://help.sap.com/docs/build-process-automation/sap-build-process-automation/use-api-keys?locale=en-US>
- **SAP Help: Configure an API Trigger to Start a Process:** <https://help.sap.com/docs/build-process-automation/sap-build-process-automation/configure-and-test-api-call-to-trigger-process?locale=en-US>